



**Year 12 HSC Assessment Task 1
Mathematics Extension 2
Week 9A Monday 1st Dec 2014**

Name: _____

Teacher: _____

- Attempt **ALL** questions.
 - Marks may be deducted for insufficient or illegible work.
 - Only Board approved calculators (**excluding** graphic calculators) may be used.
 - Total possible mark is **32**.
 - Begin each question on a new sheet of paper.
 - **TIME ALLOWED: 40 minutes plus 2 minutes reading time.**
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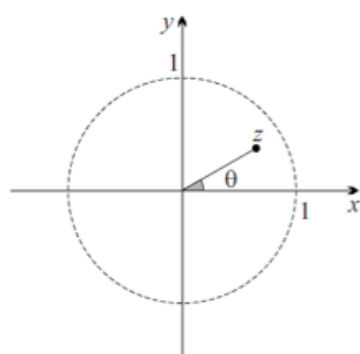
SECTION I – Multiple Choice (4 marks)

- 1 Which expression is a correct factorization of $z^3 - i$?
- (A) $(z - i)(z^2 + iz + 1)$
- (B) $(z + i)(z^2 - iz - 1)$
- (C) $(z + 1)(z - i)^2$
- (D) $(z + 1)^3$
- 2 The complex number w is a root of the equation $z^3 + 1 = 0$. Which of the following is FALSE.
- (A) $\overline{w}$ is also a root
- (B) $w^2 + 1 - w = 0$
- (C) $\frac{1}{w}$ is also a root.
- (D) $(w - 1)^2 = -1$

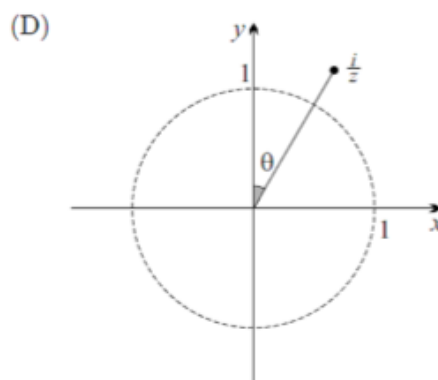
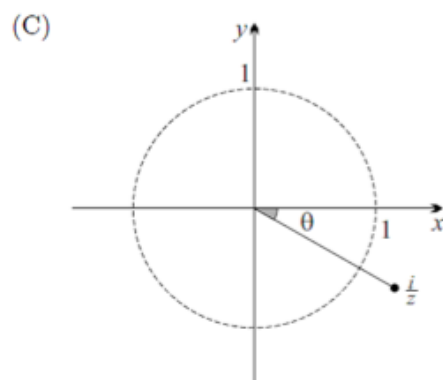
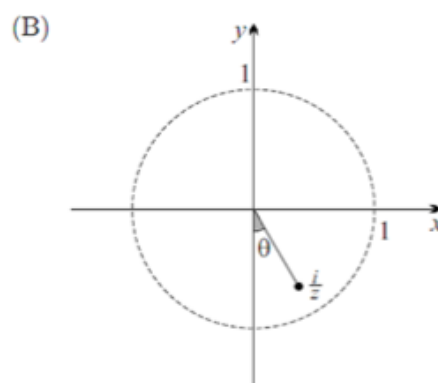
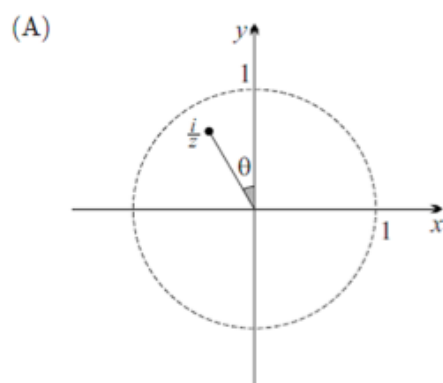
3 By de Moivre's theorem, the value of $(1+i)^{10}$ is:

- (A) purely real
- (B) purely imaginary
- (C) a real multiple of $(1+i)$
- (D) an imaginary multiple of $(1+i)$

4



The Argand diagram above shows the complex number z . By considering the modulus and argument, which diagram below best represents the complex number $\frac{i}{z}$?



SECTION II

Question 5 (14 marks) Use a SEPARATE writing booklet.

Marks

- (a) Consider the complex numbers $z = 2 + 3i$ and $w = -1 + 2i$.

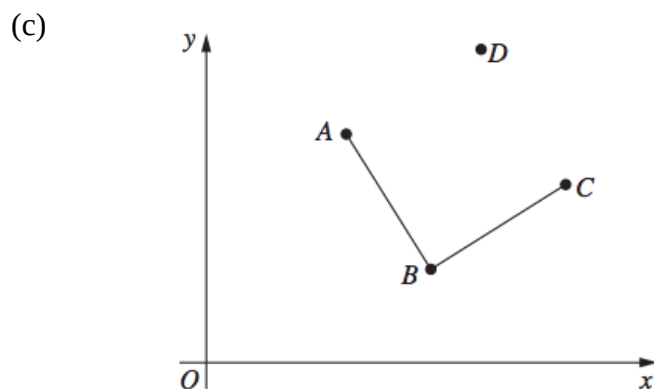
Express each of the following in the form $x + iy$, where x and y are real numbers.

(i) $z + \overline{w}$ 1

(ii) z^2 1

(iii) $\frac{z}{w}$ 1

- (b) Find the square roots of $21 - 20i$ in the form $a + ib$ where a and b are real numbers. 3



In the diagram the vertices of a triangle ABC are represented by the complex numbers z_1, z_2 and z_3 respectively. The triangle is isosceles and right-angled at B .

(i) Show that $(z_1 - z_2)^2 = -(z_3 - z_2)^2$. 2

- (ii) Suppose D is the point such that $ABCD$ is a square. Find the complex number, expressed in terms of z_1, z_2 and z_3 that represents D . 1

- (d) Use de Moivre's Theorem, or otherwise, show that for every positive integer n 2

$$(1+i)^n + (1-i)^n = 2(\sqrt{2})^n \cos \frac{n\rho}{4}.$$

(Do not use Mathematical Induction)

- (e) Prove using the principle of Mathematical induction that, 3

$$\text{If } T_1 = 1, T_2 = 5 \text{ and } T_n = 5T_{n-1} - 6T_{n-2} \text{ for } n \geq 3,$$

show $T_n = 3^n - 2^n$ for $n \geq 1$

End of Question 5

Question 6 (14 marks) Use a SEPARATE writing booklet.

Marks

- (a) Sketch the region in the Argand diagram where 3

$$|z - 1 - i| \leq 1 \text{ and } -\frac{\rho}{4} \leq \text{Arg}(z - 1 - i) \leq \frac{\rho}{2}$$

- (b) Find the fourth roots of $-8 + 8i\sqrt{3}$ and plot the roots on an Argand diagram. 3

- (c) The inequality $x > \ln(1+x)$ holds for all real $x > 0$. 3
(Do NOT prove this.)

Use this result and the method of mathematical induction to prove that for all positive integers n ,

$$\frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} > \ln(n+1).$$

- (d) Let the points $A_1, A_2, \dots, A_n$ represent the n th roots of unity, $w_1, w_2, \dots, w_n$, and suppose P represents any complex number z such that $|z| = 1$.

- (i) Prove that $w_1 + w_2 + \dots + w_n = 0$. 2

- (ii) Show that $(PA_i)^2 = (z - w_i)(\bar{z} - \bar{w}_i)$ 1
for $i = 1, 2, \dots, n$.

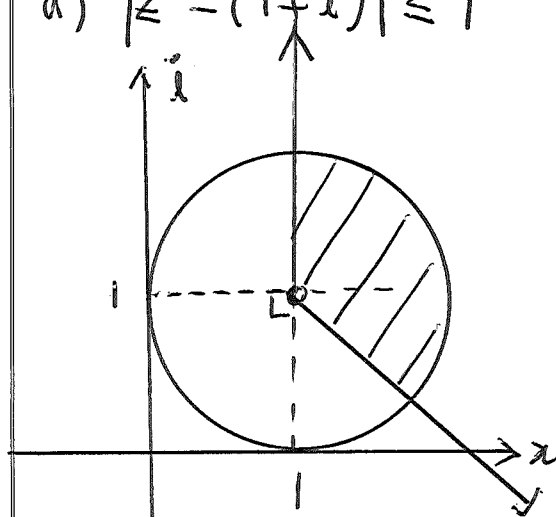
- (iii) Prove that $\sum_{i=1}^n (PA_i)^2 = 2n$ 2

End of paper

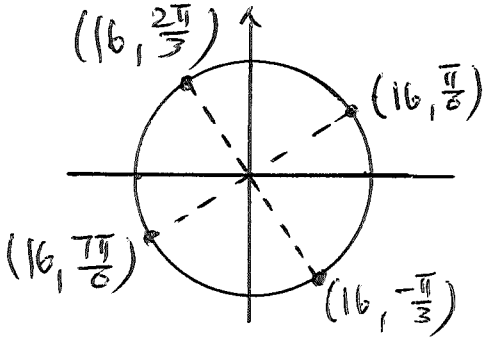
2014 Year 12 Mathematics Extension 2 HSC Task 1 Solutions

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
<u>SECTION I</u> 1. $z^3 + (\bar{z})^3$ B $= (z + \bar{z})(z^2 - \bar{z}z - i)$ 2. $z^3 + 1$ $= (z + 1)(z^2 - z + 1)$ So $w^3 + 1 = (w + 1)(w^2 - w + 1)$ Now $(w - 1)^2$ $= w^2 - 2w + 1$ $= -w$ D 3. $(1 + i)^{10} = (\sqrt{2} \operatorname{cis} \frac{\pi}{4})^{10}$ $= 32 \operatorname{cis} \frac{5\pi}{2}$ $= 32 \operatorname{cis} \frac{\pi}{2}$ $= 32i$ B. 4. $\operatorname{Arg} \left(\frac{1}{z} \right) = -\arg z$ $\operatorname{Arg} \left(\frac{\bar{z}}{z} \right) = \frac{\pi}{2} - \arg z$ D.		<u>SECTION II</u> <u>Question 5</u> a) (i) $2 - 3i + (-1 - 2i)$ $= 1 - i$ ✓ (ii) $z^2 = (2 + 3i)^2$ $= -5 + 12i$ ✓ (iii) $\frac{z}{w} = \frac{2 + 3i}{-1 + 2i} \cdot \frac{-1 - 2i}{-1 - 2i}$ $= \frac{4 - 7i}{5}$ $= \frac{4}{5} - \frac{7i}{5}$ ✓ b) Let $z^2 = 21 - 20i$ With $z = a + bi$ So $(a + bi)^2 = 21 - 20i$ $a^2 - b^2 = 21$ $ab = -10$ Also $a^2 + b^2 = (a + bi)^2$ $= 21 - 20i$ $= 29$ Solving simultaneously, $a = 5 \quad b = -2$ $a = -5 \quad b = 2$	

$$z = \pm (5 - 2i)$$

	Comments	Suggested Solution (s)	Comments
(c)(i) $\overrightarrow{BA} = z_1 - z_2$ $\& \overrightarrow{BC} = z_3 - z_2$ Rotating $\overrightarrow{BC}$ anticlockwise by 90° gives $\overrightarrow{BA}$. Hence $z_1 - z_2 = i(z_3 - z_2)$ Squaring both sides gives $(z_1 - z_2)^2 = -(z_3 - z_2)^2$ (ii) $\overrightarrow{OD} = \overrightarrow{OA} + \overrightarrow{AD}$ But $\overrightarrow{AD} = \overrightarrow{BC}$ $\therefore \overrightarrow{OD} = \overrightarrow{OA} + \overrightarrow{BC}$ $= z_1 + (z_3 - z_2)$ So D is represented by the complex number $z_1 - z_2 + z_3$. (d) To show $(1+i)^n + (1-i)^n$ $= 2(\sqrt{2})^n \cos \frac{n\pi}{4}$ LHS = $(\sqrt{2} \operatorname{cis} \frac{\pi}{4})^n + (\sqrt{2} \operatorname{cis} (-\frac{\pi}{4}))^n$ $= 2^{\frac{n}{2}} \left[\cos \frac{n\pi}{4} + \cos \left(-\frac{n\pi}{4}\right) \right]$ $+ i \left(\sin \frac{n\pi}{4} + \sin \left(-\frac{n\pi}{4}\right) \right)$ $= 2 \cdot 2^{\frac{n}{2}} \cos \frac{n\pi}{4}$ $= 2(\sqrt{2})^n \cos \frac{n\pi}{4}$	✓	(e) $T_1 = 1, T_2 = 5$ $T_{k-1} = 3^{k-1} - 2^{k-1}$ $T_k = 3^k - 2^k$ RTP $T_{k+1} = 5T_k - 6T_{k-1}$ Now $5(3^k - 2^k) - 6(3^{k-1} - 2^{k-1})$ $= 5 \times 3^k - 6 \times 3^{k-1} - 5 \times 2^k + 6 \times 2^{k-1}$ $= 5 \times 3^k - 2 \times 3^k - 5 \times 2^k + 3 \times 2^k$ $= 3 \times 3^k - 2 \times 2^k$ $= 3^{k+1} - 2^{k+1}$ Question 6 a) $z - (1+i) \leq 1$ 	

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	Comments	Suggested Solution (s)	Comments
b) $z_k = 16 \operatorname{cis} \left(\frac{2\pi}{3} + \frac{2k\pi}{4} \right)$ $k = 0, 1, 2, 3$ $= 16 \operatorname{cis} \left(\frac{2\pi + 6\pi k}{12} \right)$ $= 16 \operatorname{cis} \left(\frac{\pi + 3k\pi}{6} \right)$ $= 16 \operatorname{cis} (3k+1) \frac{\pi}{6}$ $z_0 = 16 \operatorname{cis} \frac{\pi}{6}$ $z_1 = 16 \operatorname{cis} \frac{4\pi}{6} = 16 \operatorname{cis} \frac{2\pi}{3}$ $z_2 = 16 \operatorname{cis} \frac{7\pi}{6}$ $z_3 = 16 \operatorname{cis} \left(\frac{10\pi}{6} \right)$ $= 16 \operatorname{cis} \frac{5\pi}{3}$ $= 16 \operatorname{cis} \left(-\frac{\pi}{3} \right)$ 		c) Let $p(n)$ be the given proposition $\frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} > \ln(n+1)$ $p(1)$ is true since $\frac{1}{1} > \ln(1+1)$ (Note: $e > 2 \Rightarrow \ln e > \ln 2$ $\Rightarrow 1 > \ln 2$) Assume $p(k)$ is true for some positive integer k. i.e) $\frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{k} > \ln(k+1)$ Prove true for $p(k+1)$ $\frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{k} + \frac{1}{k+1}$ $> \ln(k+1) + \frac{1}{k+1}$ (using the assumption) $> \ln(k+1) + \ln \left(1 + \frac{1}{k+1} \right)$ (using the inequality given) $= \ln \left((k+1) \left(1 + \frac{1}{k+1} \right) \right)$ $= \ln(k+1 + 1)$	

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	Comments	Suggested Solution (s)	Comments
d) (i) $w^n = 1$ $(w \text{ is an } n^{\text{th}} \text{ root of unity})$ Consider $w + w^2 + \dots + w^n = 0$ $S_n = \frac{a(1-r^n)}{1-r}$ $S_n = \frac{w(1-w^n)}{1-w}$ but $w^n = 1$ $= \frac{w(1-1)}{1-w}$ $= 0$ ii) $PA_i = z - w_i$ $PA_i ^2 = (z - w_i)^2$ $= (z - w_i) \overline{(z - w_i)}$ since $z ^2 = z \cdot \bar{z}$ $= (z - w_i)(\bar{z} - \bar{w}_i)$		(iii) $\sum_{i=1}^n PA_i ^2 = \sum_{i=1}^n (z - w_i)(\bar{z} - \bar{w}_i)$ $= \sum_{i=1}^n (z\bar{z} - z\bar{w}_i - w_i\bar{z} + w_i\bar{w}_i)$ $= \sum_{i=1}^n (2 - z\bar{w}_i - w_i\bar{z})$ since $z\bar{z} = z ^2 = 1$ $w\bar{w} = w ^2 = 1$ $= 2n - z \sum_{i=1}^n \bar{w}_i - \bar{z} \sum_{i=1}^n w_i$ $= 2n - z \cdot 0 - \bar{z} \cdot 0$ (by (i)) $= 2n$	